

A FIELD MANUAL FOR BIM COORDINATORS

The IFC Delivery Handbook

How to check, prove and hand over IFC models that get accepted the first time

44

checks, documented
one by one

4

authoring tools,
step by step

7

clauses and templates,
ready to paste

The IFC Delivery Handbook

How to check, prove and hand over IFC models that get accepted the first time

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The check reference in chapter 4, the score weights in chapter 3 and the tool-by-tool remediation steps are generated directly from the source of the validator at www.ifcvieweronline.eu, so this handbook always documents the checks that actually run.

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How to use this handbook

This is a working document, not a textbook. It is written for the person who has to press **Upload** on a common data environment and put their name next to the file.

There are three ways to read it, and they are all legitimate:

1 You have a delivery due this week

Go straight to **Appendix A — the pre-delivery checklist**, run it, and come back for the chapters behind whatever failed.

2 You keep receiving models you cannot use

Read **chapter 7**. Your problem is not technical, it is contractual: nobody wrote down what "acceptable" means. Chapter 7 gives you the clauses to paste into the BEP.

3 You want to build a repeatable quality process

Read it in order. Chapters 1–3 build the vocabulary, chapter 4 is the reference you will return to, chapters 5–7 are the process.

What this handbook assumes

That you can export an IFC file from your authoring tool, and that you have opened one before. It does not assume you know the IFC schema, ISO 19650 clause numbers, or what `IfcRelContainedInSpatialStructure` means — where those matter, they are explained in place.

What this handbook is not

- It is not a modelling guide. How you model a wall is your business; how that wall arrives in the IFC is ours.
- It is not a substitute for your project's Exchange Information Requirements (EIR). It is the layer *underneath* them — the checks that any EIR, in any language, ends up depending on.
- It is not vendor-neutral theatre. The 44 checks in chapter 4 are the exact checks the free validator at ifcvieweronline.eu runs, quoted from its source. You can reproduce every one of them yourself, with any tool that implements the same rule, or by hand.

On the checks in chapter 4

The rule reference is generated from the validator's own rule table at build time — identifier, category, default severity, referenced standard and the tool-by-tool remediation text. That is deliberate: a handbook that drifts away from the tool it documents is worse than no handbook.

Why good models get rejected

01

Almost nobody delivers a bad model on purpose. Models get rejected because the sender and the receiver are measuring different things — and neither of them wrote down what they were measuring.

The acceptance gap

Here is the sequence, and you have almost certainly lived it. An architect exports an IFC from a model that is, from the inside, perfectly good: the geometry is right, the levels are right, the sheets print. It goes to the CDE. Three days later the MEP coordinator replies that half the elements are not on any storey, the space names are empty, and their software is putting the building 400 kilometres from site.

Nobody is lying. The model *is* good — inside the authoring tool. The IFC *is* broken — outside it. The difference between those two sentences is the acceptance gap, and it exists because of one structural fact:

An IFC export is a translation, and translations lose things that the original author never had to think about.

Inside Revit or ArchiCAD or Tekla, a wall belongs to a level because the software says so. The relationship is implicit, enforced by the interface, invisible to the modeller. In IFC that relationship is an explicit object — `IfcRelContainedInSpatialStructure` — that the exporter has to write down. If the exporter cannot work out which storey the wall belongs to, it writes nothing. The wall still renders. It just belongs to nothing, and every downstream tool that groups by storey silently drops it.

That is the shape of nearly every IFC delivery failure: **the thing that was implicit did not survive being made explicit.**

Three failure modes, in order of cost

FAILURE MODE	WHAT IT LOOKS LIKE	WHO PAYS
Structural	Duplicate or malformed GUIDs, broken aggregation, elements outside the spatial tree, circular references	Everyone downstream. The file cannot be reliably merged, compared or federated at all.
Informational	Missing property sets, empty names, no classification, no materials, spaces without area	The quantity surveyor, the FM team, the COBie deliverable. Usually discovered months later.
Contextual	Wrong units, huge coordinate offsets, storeys out of order, missing project metadata	The coordinator, immediately and loudly. Cheapest to fix, most visible when missed.

The categories used throughout this handbook map onto these three modes: schema and spatial faults are structural; quality, LOD, classification and MEP faults are informational; ISO 19650 and coordinate faults are contextual.

Structural faults are the expensive ones because they are *silent*. A duplicate GUID does not crash anything. It quietly makes two elements indistinguishable, so the clash report references the wrong one, the issue tracker attaches a comment to the wrong wall, and the change log between revision A and revision B becomes fiction.

Why "it opens fine" is not a check

The single most common quality assurance procedure in the industry is: export the IFC, open it in a viewer, rotate it, see the building, ship it. This catches exactly one class of problem — catastrophic geometry loss — and misses everything in the table above.

It fails because rendering has almost no requirements. A viewer needs shapes and placements. It does not need names, GUID uniqueness, a spatial hierarchy, materials, classifications, property sets, or units that make sense. You can render a model that is, as an *information container*, close to worthless.

The visual check paradox

The better your geometry, the more confident everyone becomes, and the less likely anyone is to check the data. Beautiful models are rejected more often than ugly ones, because ugly ones get inspected.

What a real check costs

Do the arithmetic yourself, with your own numbers. A validation run on a typical discipline model takes under a minute. Fixing what it finds — assuming the faults are systematic, which they usually are, because they come from export settings rather than from individual elements — takes between ten minutes and an afternoon.

Now price the alternative: one rejected delivery costs a coordination cycle. If your project federates weekly, a rejection costs a week. Multiply the week by the number of people waiting on that model. That is the entire business case, and it does not need an industry statistic to be convincing — you already know both numbers for your own project.

The three questions this handbook answers

1 Is this file structurally sound?

Chapters 3 and 4. Forty-four checks, grouped into eight categories, with a single 0–100 score so you can say "yes" or "no" without reading a report.

2 Can I prove it?

Chapters 5 and 6. A delivery is an assertion. An assertion without evidence is an opinion, and opinions get re-litigated at every gateway. The evidence pack turns your check into an artefact somebody else can verify.

3 Who agreed to what?

Chapter 7. The clauses that turn "the model was rubbish" into "clause 4.3 of the BEP was not met", which is a conversation with an ending.

What “delivered” actually means

02

ISO 19650 gives the industry a shared vocabulary for handing information over. Most teams have adopted the words — WIP, Shared, Published, Archived — without adopting the thing the words were invented to protect: a *state change that somebody is accountable for*.

The container, not the model

ISO 19650 does not talk about models. It talks about **information containers** — a named, versioned, status-carrying unit of information. Your IFC file is one container. The drawing set is another. The COBie spreadsheet is another.

This matters more than it sounds. A container has three properties that a "model" does not:

- **Identity** — a name that follows an agreed convention, so it can be found and its origin understood without opening it.
- **Revision** — a version marker, so "the latest one" is a fact rather than an argument about email timestamps.
- **Status** — suitability. What the receiver is allowed to do with it: look at it, coordinate against it, build from it.

Two of the three live outside the file, in the CDE. One of them — identity — can and should live *inside* it too, which is why a filename check appears among the 44.

The four states, in the only terms that matter

STATE	QUESTION IT ANSWERS	WHAT MUST BE TRUE
Work in progress	Is this mine?	Nothing. It is yours, it can be broken, nobody else may reference it.
Shared	May a colleague coordinate against this?	It must be structurally sound and honestly labelled. This is where the pre-flight check belongs.
Published	May somebody make a decision — or a purchase — from this?	It must meet the EIR in full, and the evidence must exist.
Archived	What did we say, and when?	It must be immutable and reproducible. A verifiable check record belongs here.

The one process change worth making

Run the check at the **Work in progress** → **Shared** boundary, not at Published. By the time something is published, three disciplines have already coordinated against it, and a structural fault means re-doing their work, not yours.

Suitability is a promise about data, not about geometry

When you mark a container as suitable for coordination, you are promising that a coordinator can federate it, filter it by storey, select an element and get a stable identifier back that will still mean the same element next week. Every one of those promises is a data promise. None of them is about how the building looks.

This is why the checks in chapter 4 are weighted the way they are: a broken spatial hierarchy outranks a missing material, because the first one breaks federation and the second one only degrades a schedule.

Level of information need

The other half of ISO 19650 that gets quoted and not used: information should be delivered to the level of detail the *purpose* requires, and no further. Over-delivery is a defect, not generosity — it costs modelling time, it inflates files, and it creates data nobody validates and everybody trusts.

In practice, three things define the level of information need for an IFC delivery:

1 Geometric detail

How the element is represented. Rarely the constraint in coordination; frequently the constraint in file size (chapter 5).

2 Alphanumeric detail

Which property sets and which properties. This is what the LOD checks in chapter 4 test — do the elements carry the property sets their declared level implies?

3 Documentation

What travels alongside: classification references, materials, system assignments, the evidence pack of chapter 6.

A model at a high geometric level with empty property sets is not a high level of information need. It is a picture.

The handover asymmetry

One last structural point before the checks. The sender knows what they meant. The receiver knows only what arrived. Every ambiguity in an IFC file is resolved by the receiver — usually by guessing, occasionally by asking, never by reading your mind.

So the target is not "a model I can defend". The target is **a model that needs no defending**, because everything a receiver would have to guess about has been made explicit in the file. That is the whole design brief of the next two chapters.

The pre-flight check

03

A pre-flight check has to answer one question in one glance: **do I send this, yes or no?** Everything else — the issue list, the element identifiers, the export settings to change — is the follow-up. This chapter builds that answer out of three parts: eight categories, one score, and an honesty mechanism most validators do not have.

Eight categories

Every check in chapter 4 belongs to exactly one category. The categories exist so you can triage a report without reading it — a run with twelve *schema* findings is a different emergency from a run with twelve *classification* findings.

● IFC Schema

9 checks · 5 at error

Faults in the file itself: identifiers, references and structural integrity. These break tools rather than degrade information, which is why they carry the heaviest weights.

● Spatial structure

11 checks · 3 at error

The project → site → building → storey → space hierarchy every element should hang from. Break it and filtering, scheduling and federation quietly degrade.

● Data quality

13 checks · 1 at error

Information that is present but empty, missing or meaningless. Rarely fatal on its own, cumulatively the difference between a model and a picture.

● LOD / LOIN

3 checks

Whether elements carry the alphanumeric detail their declared level of information need implies — property sets, quantities, material layers.

● ISO 19650

3 checks

Project and author metadata, and the naming convention that lets an information container be identified without opening it.

● Classification

1 check

Whether elements reference an agreed classification system, which is what makes cost, procurement and asset data possible.

● MEP

2 checks

System assignment and connectivity for services elements — the difference between pipes and a network.

● Clash detection

2 checks

Geometric interference between elements. Included as a coarse pre-flight sweep, not as a substitute for a coordination clash workflow.

One number: the Health Score

A score is a compression of a report. It is useful precisely because it throws information away — it lets you set a gate in a contract, compare revision A to revision B, and tell a client something true in one sentence.

It is also the part of any validator that is most often hand-waved, so here is the exact model used throughout this handbook, in full:

Step 1 — every finding carries a weight

The weight depends on the category of the rule and the severity of the finding. Structural categories are weighted harder than informational ones, for the reason given in chapter 2.

CATEGORY	ERROR	WARNING	INFO
IFC Schema	5	1.5	0.3
Spatial structure	4	1.2	0.2
Data quality	3	1	0.2
LOD / LOIN	3	1	0.2
ISO 19650	3	0.8	0.1
Classification	2	0.6	0.1
MEP	2	0.8	0.1
Clash detection	4	1.5	0.2

Points deducted for the first occurrence of a finding. Generated from the validator's own weight table.

Step 2 — repeats cost less than the first offence

This is the part that makes a score usable on real models. A naive validator multiplies weight by occurrence count: five thousand elements missing a material becomes a

penalty of five thousand, every real model scores zero, and the number tells you nothing.

But five thousand elements missing a material is not five thousand problems. It is **one** problem — an export setting — that touched five thousand elements. So the penalty grows logarithmically:

```
penalty(rule) = weight(category, severity) × (1 + ln(occurrences))
```

The first occurrence costs the full weight. Ten occurrences cost about 3.3×, not 10×. A thousand cost about 7.9×, not 1000×. The score therefore ranks *distinct defects*, which is what you actually fix.

Step 3 — subtract from 100 and clamp

```
score = max(0, round(100 - ∑ penalty(rule)))
```

Two consequences worth internalising. First, the score is bounded below, so a catastrophic model and a merely awful one both read 0 — below about 40 the number stops discriminating and you should read the report instead. Second, because the penalty is per-rule and not per-element, fixing one systematic defect can move the score by ten points in a single export setting.

Reading the number

SCORE	READING	ACTION
85 – 100	Deliverable. Minor observations at most.	Ship it. Note any waived findings in the transmittal.
70 – 84	Deliverable with observations. Nothing structural, but something is missing.	Ship only if the missing information is outside the current level of information need — and say so explicitly.
Below 70	Not deliverable.	Fix, re-run, then ship. Sending it anyway transfers your problem to somebody who cannot fix it.

These are the same thresholds the viewer uses for its client-facing badge: 85 for a green “verified” reading, 70 for amber. Below 70 the honest answer to “is this ready?” is no.

Do not turn the score into a target

The moment a score becomes a KPI, somebody optimises it by disabling rules. A score is only meaningful alongside the rule set that produced it — which is why the delivery evidence in chapter 6 records both.

Coverage: the check nobody runs on the checker

Here is a failure mode specific to automated validation, and it is worse than any defect in this book: **a rule that did not run looks exactly like a rule that passed**. Both produce zero findings.

A large file times out, a worker crashes, a geometry-dependent rule silently gives up — and the report comes back clean. The score is 100. You ship, confident, having checked nothing.

The fix is to make every run report three states per rule rather than two:

STATE	MEANING	WHAT TO DO
ok	The rule ran to completion.	Trust the finding count, including zero.
not-run	The rule was attempted but produced no outcome — usually a timeout or a cancelled run.	Re-run, or narrow the scope. Do not treat as a pass.
failed	The rule errored.	Report it. A score computed over failed rules is not comparable to one computed over a clean run.

Whatever tool you use, ask it one question: *how do I know all the checks ran?* If it cannot answer, its clean reports mean less than you think.

Why this handbook keeps saying “whatever tool you use”

Every check in chapter 4 is a statement about an IFC file, not about a product. They are reproducible in any checker, and several are reproducible with a text editor and patience. The tool this handbook comes from is free and runs in the browser, which makes it convenient — it does not make it necessary.

Severity is a project decision

Each rule ships with a default severity — error, warning or info — and every one of those defaults is arguable on a specific project. Missing classification is an error on an asset-management-driven public project and an irrelevance on a fast-track fit-out.

So treat the defaults in chapter 4 as a starting position, and record your deviations in the BEP (chapter 7) rather than in somebody's head. Two things follow: the score becomes comparable across your own revisions, and "we don't care about that one" becomes a decision with a date and an author.

The 44-check reference

The rest of this chapter is the reference. Every check the validator runs, in the order it runs them, grouped by category: what it tests, why it matters, and how to fix it in the four authoring tools that produce most of the industry's IFC.

Each entry carries its identifier, its category, its default severity and the standard it derives from. The identifiers are stable — quote them in a BEP or an issue and they will still mean the same check next year.

How to use this chapter

Do not read it end to end. Run a check, take the rule identifiers it reports, look them up here, and fix the export setting rather than the elements. Roughly nine findings in ten are a setting, not a modelling mistake.

● IFC Schema

9 CHECKS

Faults in the file itself: identifiers, references and structural integrity. These break tools rather than degrade information, which is why they carry the heaviest weights.

01 Duplicate GUID

RULE_DUPLICATE_GUID

ERROR

AUTO-FIXABLE

Two or more elements share the same GlobalId

CATEGORY IFC Schema REFERENCE IFC schema / ISO 10303-21

THE FIX

Every element must have a unique GlobalId. This tool can auto-fix duplicates (click Apply fix); to prevent it at the source, fix the export workflow below.

Revit

Duplicate GUIDs usually come from copy/pasting elements between models or across linked files. Avoid duplicating elements across exported models and re-export from a clean copy. For grouped or mirrored elements sharing an IfcGUID parameter, clear that parameter so Revit regenerates a unique value.

Tekla Structures

Duplicate GUIDs come from copied objects across models. Re-export from the source model — Tekla assigns a unique GUID per object at creation.

ArchiCAD

Duplicate GUIDs typically arise from copying elements between projects or merging modules. Re-generate unique IDs (Design ► Element ID Manager) and avoid copying elements across files without regenerating GlobalIds.

Allplan

Duplicate GUIDs come from copying objects between documents. Recreate or re-export the affected objects so Allplan regenerates unique GlobalIds.

02 Invalid GUID format

RULE_INVALID_GUID_FORMAT

ERROR

AUTO-FIXABLE

GlobalId does not follow IFC base-64 encoding

CATEGORY IFC Schema REFERENCE IFC schema

THE FIX

GlobalId must be a 22-character IFC base-64 string. This tool can auto-fix the format; at the source, avoid post-processing that rewrites GUIDs.

Revit

Revit writes compliant IfcGUIDs by default. Invalid formats usually come from third-party scripts or a manually edited IfcGUID parameter — clear the parameter so Revit regenerates a valid 22-character GUID on export.

Tekla Structures

Tekla writes valid GUIDs natively; invalid values typically come from interop scripts — re-export from the native model.

ArchiCAD

ARCHICAD generates compliant GlobalIds. Invalid values usually come from external edits or add-ons; regenerate IDs or re-export without the offending add-on.

Allplan

Allplan generates valid GlobalIds; if invalid, recreate or re-export the affected objects.

03 Broken aggregate

RULE_BROKEN_AGGREGATE

ERROR

IfcRelAggregates points to a non-existent entity

CATEGORY IFC Schema REFERENCE IFC schema / ISO 10303-21

THE FIX

Fix the broken aggregation relationship — this is almost always an export/interop artefact, so re-export from the authoring tool.

Revit

Re-export with an up-to-date IFC exporter. If it persists, audit the model (Manage ► Purge Unused) and check for corrupt groups or assemblies.

ArchiCAD

Re-export with the latest ARCHICAD IFC add-on; run a model check if corruption persists.

Tekla Structures

Re-export from Tekla — broken aggregates indicate an interop fault, not a modelling error.

Allplan

Re-export from Allplan with an up-to-date IFC interface.

04 Circular reference

RULE_CIRCULAR_REFERENCE

ERROR

Aggregate or containment relationship forms a cycle

CATEGORY IFC Schema REFERENCE IFC schema

THE FIX

Remove the circular relationship — an element cannot be its own ancestor. This is an export/interop artefact; re-export from a clean copy.

Revit

Re-export with an up-to-date IFC exporter from a clean copy; if it persists, audit and purge the model.

ArchiCAD

Re-export with the latest IFC add-on; run a model check to find the offending relationship.

Tekla Structures

Re-export from the native model — Tekla does not normally create reference cycles.

Allplan

Re-export from Allplan; recreate the affected objects if the cycle persists.

File uses an outdated or unrecognised IFC schema

CATEGORY IFC Schema REFERENCE ISO 10303-21

THE FIX

Export to a current IFC schema (IFC4 / IFC4.3) unless the recipient explicitly requires IFC2x3.

Revit

In the IFC export dialog, set File Version to IFC4 (e.g. Reference View or Design Transfer View) instead of IFC2x3.

ArchiCAD

In the IFC Translator, choose an IFC4-based export preset instead of IFC2x3.

Tekla Structures

In the IFC export, select the IFC4 export type rather than IFC2x3.

Allplan

Select IFC4 (or IFC4.3) as the export schema in the IFC interface settings.

File contains no IfcProject entity

CATEGORY IFC Schema REFERENCE IFC schema

THE FIX

Every IFC must contain exactly one IfcProject. A missing project means a broken export — re-export the full model.

Revit

Revit always writes an IfcProject. A missing one indicates a corrupt or partial export — re-export the full model rather than an isolated selection.

ArchiCAD

Re-export the project; export the model, not an isolated element set that drops the project root.

Tekla Structures

Re-export the full model so the IfcProject root is written.

Allplan

Re-export from the project so the IfcProject entity is included.

RULE_FILE_DESCRIPTION_MISSING

IFC STEP header FILE_DESCRIPTION is empty

CATEGORY IFC Schema REFERENCE ISO 10303-21 §8.2.1

THE FIX

Set the file description (usually the MVD / view definition) in the export options — it is part of the STEP header metadata.

Revit

FILE_DESCRIPTION is set from the chosen MVD (e.g. Reference View). Selecting a proper export setup in the IFC dialog populates it.

ArchiCAD

The IFC Translator's MVD selection populates FILE_DESCRIPTION; choose a defined export preset.

Tekla Structures

The export type / MVD sets FILE_DESCRIPTION; choose a defined IFC export configuration.

Allplan

Select a defined IFC export preset so the file description / MVD is written.

RULE_FILE_AUTHOR_MISSING

IFC STEP header FILE_NAME has no author specified

CATEGORY IFC Schema REFERENCE ISO 10303-21 §8.2.2

THE FIX

Fill in the author and organisation in the export or project info — required for traceability (ISO 19650).

Revit

Set the author in the IFC export setup (Modify setup) or in Manage ► Project Information; this populates the STEP FILE_NAME author field.

ArchiCAD

Set author and company in File ► Info ► Project Info and the IFC Translator so they are written to the header.

Tekla Structures

Set the author/organisation in the IFC export advanced settings.

Allplan

Set author/organisation in the project info / IFC export settings.

Model LENGTHUNIT is imperial (foot/inch) — IFC interchange requires SI metric units

CATEGORY IFC Schema REFERENCE ISO 16739

THE FIX

Export in SI metric (millimetres/metres) — imperial length units break interoperability with most IFC/BIM tools.

Revit

Revit's internal units are imperial but IFC should be metric. Set Project Units to metric (or confirm the IFC export uses SI/metric) so the file's IFCSIUNIT is metre-based.

ArchiCAD

Set the project Working Units (and Calculation Units) to metric so the IFC schema exports SI length units.

Tekla Structures

Switch the environment/role or export settings to metric so the IFC LENGTHUNIT is SI (mm/m).

Allplan

Set the length units to metric in the project options so the IFC export uses SI units.

● Spatial structure

11 CHECKS

The project → site → building → storey → space hierarchy every element should hang from.
Break it and filtering, scheduling and federation quietly degrade.

10

Orphan element

ERROR

RULE_ORPHAN_ELEMENT

Physical element with no spatial container or aggregate parent

CATEGORY Spatial structure REFERENCE IFC schema

THE FIX

Place the element inside a spatial container (storey or space) so it appears in the model tree and in downstream tools.

Revit

Orphans come from elements not assigned to a Level (groups, imported geometry, unhosted elements). Assign the element to a Level so Revit exports it inside an IfcBuildingStorey.

ArchiCAD

Check the element's Home Story setting — elements with no home story export as orphans. Assign one.

Tekla Structures

Assign the part to the phase/level structure used by the IFC export so it receives a spatial container; check the export's spatial structure settings.

Allplan

Assign the element to a storey node in the building structure palette so it isn't exported orphaned.

11

Wrong container

ERROR

RULE_WRONG_CONTAINER

Element contained in a spatial structure of wrong type

CATEGORY Spatial structure REFERENCE IFC schema

THE FIX

Move the element into the correct spatial container — physical building elements belong in a storey (or space), not directly under Site or Project.

Revit

Reassign the element to a building Level. Site components and topography are fine at site scope, but building elements must sit on a Level.

ArchiCAD

Set the element's Home Story to the correct storey; avoid placing building elements at site scope.

Tekla Structures

Adjust the spatial container mapping in the IFC export so parts land in the correct storey rather than the site.

Allplan

Move the object to the correct storey node in the building structure.

Spatial structure not rooted in IfcProject

CATEGORY Spatial structure REFERENCE IFC schema

THE FIX

Ensure the spatial structure follows Project ▶ Site ▶ Building ▶ Storey. Fix it in the authoring tool's project setup before export.

Revit

Revit builds this hierarchy automatically from Project ▶ Site ▶ Building ▶ Levels. A broken hierarchy usually means missing Levels or a custom export — verify Levels exist and use the default IFC site/building assignment.

ArchiCAD

Check the hierarchy in the IFC Translator and Story Settings: stories under the building, building under the site.

Tekla Structures

Configure the full spatial hierarchy (project/site/building/storey) in the IFC export dialog so it is complete and correctly ordered.

Allplan

Define the full building structure (project/site/building/storey) in the structure palette before exporting.

Element placed directly in IfcBuilding — must be inside a storey

CATEGORY Spatial structure REFERENCE IFC best practice

THE FIX

Place the element inside a storey rather than directly under the building.

Revit

Assign the element to a Level so it exports under an IfcBuildingStorey instead of the building.

ArchiCAD

Set the element's Home Story so it isn't placed at building scope.

Tekla Structures

Map the part to a storey in the IFC export spatial settings.

Allplan

Move the object to a storey node in the building structure.

No IfcBuilding found under IfcSite

CATEGORY Spatial structure REFERENCE IFC best practice

THE FIX

Add a building to the spatial structure — define an IfcBuilding in the authoring tool's project setup.

Revit

Revit creates the building automatically; a missing one usually means a site-only custom export. Verify the project has Levels and use the default building assignment in the IFC options.

ArchiCAD

Ensure a building exists in the project hierarchy / IFC Translator and that stories sit under it.

Tekla Structures

Define the building in the IFC export spatial structure settings.

Allplan

Add a building node in the building structure palette.

IfcBuilding has no IfcBuildingStorey children

CATEGORY Spatial structure REFERENCE IFC best practice

THE FIX

Add at least one storey (level) under the building.

Revit

Create Levels in the project; Revit exports Levels as IfcBuildingStoreys. A model with no Levels exports no storeys.

ArchiCAD

Define stories via Design ► Story Settings so the building has storeys.

Tekla Structures

Define levels/storeys in the IFC export spatial settings.

Allplan

Add storey nodes under the building in the structure palette.

IfcBuildingStorey contains no elements or spaces

CATEGORY Spatial structure REFERENCE BEP interno

THE FIX

Populate the empty storey or remove it — empty storeys clutter the spatial tree and often signal mis-assigned elements.

Revit

Delete unused Levels, or check that elements meant for that Level are assigned to it (not a neighbouring Level).

ArchiCAD

Remove the unused story or reassign elements' Home Story so the storey isn't empty.

Tekla Structures

Remove the empty level from the export or reassign parts to it.

Allplan

Delete the empty storey node or reassign objects to it.

IfcBuildingStorey.Elevation is null

CATEGORY Spatial structure REFERENCE LOIN / ISO 17412

THE FIX

Give every storey a defined elevation — it is required to place levels vertically and to generate floor plans.

Revit

Levels always carry an elevation in Revit; a null usually means a custom export. Verify Levels have numeric elevations and use the default IFC level export.

ArchiCAD

Set each story's elevation in Design ► Story Settings so it isn't null.

Tekla Structures

Ensure each level has a defined elevation in the level/grid settings before export.

Allplan

Define each storey's height/elevation in the building structure so it exports a value.

IfcOpeningElement not connected to any host element via IfcRelVoidsElement

CATEGORY Spatial structure REFERENCE IFC schema

THE FIX

Re-host or delete orphan IfcOpeningElement voids — every opening must cut a host element through IfcRelVoidsElement.

Revit

Orphan openings come from deleted/edited hosts or loosely exported shaft openings. Delete stray openings and recreate the void on its host (wall/floor/roof) so the relationship exports, and re-host doors/windows if the cut was lost.

ArchiCAD

Openings must belong to a wall or slab. Remove free-standing opening objects and use the Opening tool (or door/window) anchored to the host so ArchiCAD exports IfcRelVoidsElement.

Tekla Structures

Recreate the cut/opening as a feature of its host part instead of a loose object, so the void references a host on export.

Allplan

Place openings with the wall/slab opening tools so they belong to a host; delete detached opening solids.

Two or more storeys in the same building share the same elevation value

CATEGORY Spatial structure REFERENCE IFC best practice

THE FIX

Give each IfcBuildingStorey a distinct Elevation — levels at the same height break plan generation and storey filtering.

Revit

Two levels share the same elevation. In the Levels view give each Level a unique elevation (or delete the duplicate), and export only true building storeys as levels (turn off “Building Story”/export on the others).

ArchiCAD

Open Story Settings and set a unique elevation per story; merge or delete duplicated stories at the same height.

Tekla Structures

In the level/phase list assign a unique elevation to each level used for the IFC storey structure and remove duplicates.

Allplan

In the building structure set distinct heights per storey and remove duplicate storeys that resolve to the same elevation.

IfcBuildingStorey elevations are not in ascending order within the building

CATEGORY Spatial structure REFERENCE IFC best practice

THE FIX

Order storeys so their Elevation rises from bottom to top — out-of-order levels confuse section/plan tools and reviewers.

Revit

A lower level has a higher elevation (or vice versa).
Fix the level elevations or the export order so storeys read bottom-to-top, and check basement/roof levels with negative elevations.

Tekla Structures

Reorder/renumber the levels so their elevations ascend; correct any level whose height contradicts its position.

ArchiCAD

In Story Settings fix the height of any out-of-sequence story so elevations rise with the story index.

Allplan

In the building structure reorder storeys or fix their heights so elevations increase upward.

● Data quality

13 CHECKS

Information that is present but empty, missing or meaningless. Rarely fatal on its own, cumulatively the difference between a model and a picture.

21 Empty name

RULE_EMPTY_NAME

ERROR

Element with Name = "" or null

CATEGORY Data quality REFERENCE IFC schema

THE FIX

Give the element a meaningful Name so it is identifiable in schedules, the model tree and downstream coordination.

Revit

Revit maps a family Type Name to the IFC Name on export. Open the element's Type Properties and give the type a descriptive name instead of the default (e.g. rename "Basic Wall 1"). For instance-level naming, map a shared parameter to IfcName in the IFC export mapping table.

ArchiCAD

Select the element and open the IFC Manager (right-click ► IFC Manager) or the Classification & Properties palette; set IfcRoot.Name there, or define the mapping in the IFC Translator's property settings before exporting.

Tekla Structures

The part Name field maps to the IFC Name. Open the part properties, enter a Name, and confirm the IFC export setting maps that attribute to IfcName.

Allplan

Assign an attribute that Allplan maps to IfcName via the Properties palette, or set the IfcName mapping in the IFC export configuration.

22 Empty long name

RULE_EMPTY_LONGNAME

WARNING

IfcSpace/Storey/Building with no LongName

CATEGORY Data quality REFERENCE ISO 19650-2

THE FIX

Set the LongName on spaces, storeys and the building — it carries the human-readable room/level name used in schedules and COBie.

Revit

For spaces, set the Room/Area Name (Revit maps Room Name → IfcLongName and Room Number → IfcName). For storeys, give each Level a descriptive Name. For the building, set the name in the IFC export options.

ArchiCAD

Set the Zone Name for spaces (maps to IfcLongName) and name storeys via Design ► Story Settings. Set the building long name in File ► Info ► Project Info / the IFC Translator.

Tekla Structures

Spaces and storeys are seldom authored in Tekla; where present, set the Name/UDA mapped to IfcLongName, or set storey names in the IFC export spatial settings.

Allplan

Set the room name (maps to IfcLongName), name storeys in the building structure, and set the building long name in the IFC export.

Two or more siblings share the same Name

CATEGORY Data quality REFERENCE BEP interno

THE FIX

Make sibling element names unique (or rely on type + instance number) so elements are distinguishable in schedules and coordination.

Revit

Duplicate names usually come from identical type names or marks. Use the instance Mark parameter (unique per element) or rename types, and resolve Revit's duplicate-Mark warnings.

ArchiCAD

Use the ID Manager (Document ► ID Manager) to auto-assign unique element IDs so siblings don't share a Name.

Tekla Structures

Run numbering (Drawings & reports ► Numbering) so each part gets a unique position/part mark mapped to Name.

Allplan

Assign unique attribute values (e.g. component number) via the attribute palette so siblings don't share a Name.

Name does not match the project BEP pattern

CATEGORY Data quality REFERENCE ISO 19650-2 §9.2

THE FIX

Rename elements to follow the project's BEP naming pattern (usually defined in the EIR / ISO 19650 information requirements).

Revit

Standardise type names and the parameter mapped to IfcName to match the BEP. Use a shared parameter or a Dynamo script for bulk renaming, then map it to IfcName on export.

ArchiCAD

Apply the standard via the ID Manager and align the property mapped to IfcName (Classification & Properties palette) with the BEP.

Tekla Structures

Configure the numbering series and part naming to match the BEP, then re-run numbering.

Allplan

Use attribute templates / favourites to enforce the BEP naming and map that attribute to IfcName on export.

Element has no IfcTypeObject (IfcWallType, IfcDoorType, ...)

CATEGORY Data quality REFERENCE IFC schema

THE FIX

Associate the element with a type (IfcWallType, IfcDoorType, ...) so type properties and quantities propagate.

Revit

Revit family types export as IfcTypeObjects automatically. Missing types usually mean in-place families or generic models — convert them to loadable families with defined types, and keep type export enabled in the IFC options.

ArchiCAD

Use favourites / building materials and keep "Type Product" export enabled in the IFC Translator so element types are written.

Tekla Structures

Assign a profile and material so the part exports with a type; verify the IFC export writes type objects.

Allplan

Use library objects / SmartParts with defined types and enable type export in the IFC interface.

Element missing a required Pset from the project config

CATEGORY Data quality REFERENCE LOD / LOIN / BEP

THE FIX

Add the required property set(s) defined by the project's BEP/EIR to the element before export.

Revit

Add the missing parameters and map them to the required Pset via a User Defined PropertySets file referenced in the IFC export setup.

ArchiCAD

Define the required Pset in the Property Manager, assign it to the relevant classifications, and map it in the IFC Translator.

Tekla Structures

Add the properties as UDAs and map them to the required Pset in the IFC export additional property sets.

Allplan

Create the attributes and map them to the required Pset in the IFC export configuration.

IfcPropertySingleValue has a null or empty nominal value

CATEGORY Data quality REFERENCE LOIN / ISO 17412

THE FIX

Fill in the empty property value — an empty property is treated as missing by downstream checks.

Revit

Find the parameter and enter a value (or remove the empty parameter). A schedule is the fastest way to find and fill blanks in bulk.

ArchiCAD

Use the Property Manager or an interactive schedule to find and populate empty property values.

Tekla Structures

Populate the empty UDA values via the inquire/report tools before export.

Allplan

Fill the empty attribute values via the attribute palette or a list before export.

Physical element has no IfcRelAssociatesMaterial

CATEGORY Data quality REFERENCE LOD 200+

THE FIX

Assign a material to the element so it carries material data (expected from LOD 200/300 onward).

Revit

Assign a material to the element's structure (Edit Type ► Structure, or the Material parameter). Revit exports defined materials as IfcMaterial / layer sets.

ArchiCAD

Assign a Building Material (not just a surface) to the element; ARCHICAD exports Building Materials as IfcMaterial.

Tekla Structures

Set the part material in the part properties; Tekla exports it as the associated IFC material.

Allplan

Assign a material/format attribute to the element so it exports with a material association.

RULE_PROJECT_LONGNAME_MISSING

IfcProject.LongName is empty

CATEGORY Data quality REFERENCE ISO 19650-2 §6.1

THE FIX

Set the project long name (the descriptive project title) in the authoring tool's project information.

Revit

Set Project Name / Project Issue Name in Manage
▸ Project Information and map it to
IfcProject.LongName in the IFC export setup.

ArchiCAD

Set the project name/description in File ▸ Info ▸
Project Info; the IFC Translator maps it to
IfcProject.LongName.

Tekla Structures

Set the project name in Project properties and
map it to IfcProject.LongName in the export.

Allplan

Set the project name/description in project info so
IfcProject.LongName is populated.

RULE_PROXY_OVERUSE

More than 5% of elements are IfcBuildingElementProxy — typically unconverted Revit families

CATEGORY Data quality REFERENCE IFC best practice

THE FIX

Reduce IfcBuildingElementProxy elements by mapping them to proper IFC classes — proxies carry no semantic type.

Revit

Proxies come from in-place families, generic models, or unmapped categories. Use the IFC export class mapping table to map those categories to real IFC types instead of IfcBuildingElementProxy, and convert in-place families to loadable families.

ArchiCAD

Assign proper classifications / IFC types to objects (especially Morphs and custom objects) so they don't export as proxies.

Tekla Structures

Map custom or proxy parts to the correct IFC entity in the IFC export settings.

Allplan

Assign the correct IFC type to generic objects so they aren't exported as proxies.

31 Large coordinate offset

WARNING

RULE_COORDINATE_OFFSET

Model geometry is more than 10 km from the WCS origin — causes floating-point precision errors

CATEGORY Data quality REFERENCE IFC best practice

THE FIX

Keep the model near the internal origin and georeference it properly, instead of modelling at large real-world coordinates.

Revit

Don't model far from Revit's internal origin. Use Shared Coordinates with a Survey Point / Project Base Point and export with current shared coordinates so geometry stays near origin while georeferencing is preserved.

ArchiCAD

Set the Survey Point and Project Origin; keep the model near origin and use IFC georeferencing (IfcMapConversion) instead of a large offset.

Tekla Structures

Set the base/work point and keep the model near origin; use the IFC export base point so coordinates aren't huge.

Allplan

Set a project georeferencing/base point and keep geometry near origin rather than at real-world coordinates.

32 File size anomaly

INFO

RULE_FILE_SIZE_ANOMALY

File is unusually large per element — over-detailed geometry or embedded textures likely

CATEGORY Data quality REFERENCE IFC best practice

THE FIX

Reduce file weight: lower tessellation/detail, avoid embedded textures, and export only what is needed.

Revit

Lower the level of detail for export, avoid exporting imported CAD and very high-poly families, and split disciplines. The Reference View MVD produces lighter tessellated geometry.

ArchiCAD

Reduce curve/segment resolution, avoid embedding textures, use a lean IFC Translator preset, and export only the needed elements.

Tekla Structures

Reduce the export geometry detail/representation and avoid exporting reference models unnecessarily.

Allplan

Lower the geometry resolution and avoid embedding textures in the IFC export.

IfcSpace has no NetFloorArea quantity — required for energy analysis and QS workflows

CATEGORY Data quality REFERENCE IFC best practice

THE FIX

Add area quantities to IfcSpace — export BaseQuantities so each space carries NetFloorArea/GrossFloorArea.

Revit

Rooms export as IfcSpace but quantities are missing. Enable “Export base quantities” (Pset/QTO) in the IFC export options and make sure Rooms are properly bounded/placed so areas compute.

Tekla Structures

Spaces are limited in Tekla; if required, define them and enable quantity export, or generate them in the architectural model.

ArchiCAD

Use Zones for spaces and enable Base Quantities in the IFC Translator so IfcSpace exports NetFloorArea/GrossFloorArea.

Allplan

Create Rooms (spaces) and enable IFC quantity export so IfcSpace carries area quantities.

● LOD / LOIN

3 CHECKS

Whether elements carry the alphanumeric detail their declared level of information need implies — property sets, quantities, material layers.

34 LOD Pset missing

RULE_LOD_PSET_MISSING

WARNING

Element missing required property sets for the declared LOD level

CATEGORY LOD / LOIN REFERENCE buildingSMART / LOD 300

THE FIX

Add the property sets required at the declared LOD/LOIN level (per the project's information delivery plan).

Revit

Map the LOD-required parameters to their Psets via a User Defined PropertySets file in the IFC export, and ensure the elements actually carry those parameters.

ArchiCAD

Define the LOD Psets in the Property Manager, assign them to the relevant classifications, and map them in the IFC Translator.

Tekla Structures

Add the LOD-required properties as UDAs and map them to the Psets in the export.

Allplan

Create the LOD attributes and map them to the required Psets in the IFC export.

35 LOD quantities missing

RULE_LOD_QUANTITY_MISSING

WARNING

Structural element missing IfcElementQuantity for declared LOD

CATEGORY LOD / LOIN REFERENCE ISO 17412 / LOIN

THE FIX

Enable base-quantity export so elements carry IfcElementQuantity (area/volume/length) at the declared LOD.

Revit

Enable "Export base quantities" in the IFC export options; Revit then writes IfcElementQuantity for elements.

ArchiCAD

Enable base-quantity export in the IFC Translator's settings.

Tekla Structures

Enable quantity / base-quantity export in the IFC export configuration.

Allplan

Enable base quantities in the IFC export settings.

RULE_LOD_MATERIAL_LAYER_MISSING

Wall/slab missing IfcMaterialLayerSetUsage at LOD $\geq$ 300

CATEGORY LOD / LOIN REFERENCE buildingSMART / LOD 300

THE FIX

Define layered construction on walls and slabs so they export an IfcMaterialLayerSetUsage at LOD 300+.

Revit

Define the wall/floor Type's Structure layers (Edit Type ► Structure) with materials; Revit exports compound structures as IfcMaterialLayerSet.

Tekla Structures

Tekla parts are typically single-material; for layered elements define the layers/materials so the layer set exports, or confirm this rule applies to your discipline.

ArchiCAD

Use Composite structures (not a single Building Material) for walls/slabs so layers export as an IfcMaterialLayerSet.

Allplan

Use multi-layer components so the material layer set is exported.

● ISO 19650

3 CHECKS

Project and author metadata, and the naming convention that lets an information container be identified without opening it.

37 ISO 19650 project info

WARNING

RULE_ISO19650_PROJECT_INFO

IfcProject missing LongName, Description, or ObjectType required by ISO 19650

CATEGORY ISO 19650 REFERENCE ISO 19650-2:2021 §9.2

THE FIX

Complete the project metadata (long name, description, project phase/type) required by ISO 19650 information requirements.

Revit

Fill Project Name, Description and status/phase in Manage ► Project Information and map them to the IfcProject fields in the IFC export setup.

ArchiCAD

Complete the project info in File ► Info ► Project Info and map the fields in the IFC Translator.

Tekla Structures

Complete the project properties and map them to the IfcProject fields in the export.

Allplan

Complete the project info so IfcProject carries LongName, Description and ObjectType.

38 ISO 19650 author info

INFO

RULE_ISO19650_AUTHOR_INFO

STEP header has no author or organisation for traceability

CATEGORY ISO 19650 REFERENCE ISO 19650-2:2021 §9.1

THE FIX

Add both the author and the organisation to the export so the deliverable is traceable per ISO 19650.

Revit

Set author and organisation in the IFC export setup / Project Information so both appear in the STEP header.

ArchiCAD

Set author and company in Project Info and the IFC Translator.

Tekla Structures

Set author and organisation in the IFC export advanced settings.

Allplan

Set author and organisation in the project / IFC export settings.

Filename does not follow the ISO 19650 naming convention

CATEGORY ISO 19650 REFERENCE ISO 19650-2:2021 §6.3

THE FIX

Name the export file using the ISO 19650 pattern: Project-Originator-Volume-Level-Type-Role-Number.

Revit

Revit takes the filename from the export Save dialog — name the file per the ISO 19650 pattern when exporting (or rename it afterwards).

ArchiCAD

Set the filename per the ISO 19650 pattern in the export dialog, or rename the exported file.

Tekla Structures

Name the IFC output per the ISO 19650 pattern in the export dialog.

Allplan

Name the exported file per the ISO 19650 pattern in the export dialog.

● Classification

1 CHECK

Whether elements reference an agreed classification system, which is what makes cost, procurement and asset data possible.

40

Missing classification

WARNING

RULE_MISSING_CLASSIFICATION

Physical element has no IfcRelAssociatesClassification

CATEGORY Classification REFERENCE Uniclass 2015 / ISO 19650

THE FIX

Attach a classification reference (Uniclass, OmniClass, etc.) so the element carries its standard code as IfcRelAssociatesClassification.

Revit

Use a classification add-in (e.g. the free Classification Manager for Revit) to assign a Uniclass/OmniClass code, or map a shared parameter to IfcClassificationReference in the IFC export setup. Without a mapping Revit exports no classification.

Tekla Structures

Assign the classification via a UDA or the Tekla-IFC property mapping, then map that attribute to IfcClassificationReference in the IFC export additional property sets.

ArchiCAD

Open the Classification & Properties palette, choose a classification system (built-in or imported), and assign the element a classification item. ARCHICAD exports these as IfcClassificationReference automatically.

Allplan

Assign the classification code through the object attributes and ensure the IFC export configuration maps it to IfcClassificationReference.

● MEP

2 CHECKS

System assignment and connectivity for services elements — the difference between pipes and a network.

41 MEP system missing

WARNING

RULE_MEP_SYSTEM_MISSING

MEP flow element not assigned to any IfcSystem

CATEGORY MEP REFERENCE IFC MEP domain

THE FIX

Assign MEP elements to a system so they export inside an IfcSystem — needed for system-based coordination.

Revit

Ensure ducts/pipes/equipment belong to a named Revit System; unassigned elements export with no IfcSystem. Use the System Browser to find and assign them.

ArchiCAD

Assign MEP elements to an MEP system in the MEP Modeler so they export within an IfcSystem.

Allplan

Assign MEP objects to a system/network so they export inside an IfcSystem.

42 Disconnected MEP segment

WARNING

RULE_CONNECTED_MEP

IfcPipeSegment / IfcDuctSegment has no IfcDistributionPort connection

CATEGORY MEP REFERENCE IFC best practice

THE FIX

Connect MEP segments through ports — disconnected pipes/ducts export without IfcDistributionPort relationships and break system tracing.

Revit

Disconnected ducts/pipes export without ports. Fix open connectors in the MEP model (no gaps/loose ends), keep segments joined into connected systems, and enable system/port export so IfcDistributionPort relationships are written.

ArchiCAD

Use the MEP Modeler so routes stay connected end-to-end; export MEP systems to include ports/connections.

Tekla Structures

MEP is not Tekla's domain; model connected MEP in the dedicated MEP tool so segments carry ports, then federate.

Allplan

Model MEP runs connected end-to-end (no open ends) so the IFC export writes distribution ports between segments.

● Clash detection

2 CHECKS

Geometric interference between elements. Included as a coarse pre-flight sweep, not as a substitute for a coordination clash workflow.

43 Element clash

RULE_ELEMENT_CLASH

WARNING

Two structural elements have overlapping bounding boxes

CATEGORY Clash detection REFERENCE BIM coordination

THE FIX

Resolve the geometric clash between elements in the authoring tool — move, trim, or join the conflicting elements.

Revit

Run Collaborate ► Interference Check to locate clashes, then move/trim/join the elements to resolve the overlap.

ArchiCAD

Use Design ► Collision Detection to find overlaps, then adjust the conflicting elements.

Tekla Structures

Use Manage ► Clash Check to find and resolve overlapping parts.

Allplan

Use the collision check to locate overlaps and adjust the conflicting elements.

44 MEP/structural clash

RULE_CLASH_MEP_STRUCTURAL

WARNING

MEP element bounding box overlaps a structural element

CATEGORY Clash detection REFERENCE BIM coordination

THE FIX

Resolve the MEP-vs-structure clash — reroute the MEP run or coordinate a penetration/sleeve with the structural model.

Revit

Run Interference Check between the MEP and structural categories, then reroute services or add coordinated openings/sleeves.

ArchiCAD

Use Collision Detection between MEP and structural elements, then reroute or add openings.

Tekla Structures

Run a clash check against the linked MEP reference model and add penetrations/openings where needed.

Allplan

Use collision check between MEP and structure and reroute or add openings.

The delivery workflow

05

Seven steps, in order, repeatable by somebody who has never read the rest of this book. The point of writing a workflow down is not that it is clever — it is that it survives the week you are on holiday.

The seven steps

1 Export with a named, saved configuration

Not the default. Not "whatever was there last time". A configuration with a name, stored in the project template, referenced by name in the BEP. Most of chapter 4 is fixed here, once, for everybody.

2 Check the file before it goes anywhere

Before the CDE, before the email, before the WeTransfer link. The check takes under a minute; the recall email takes a day and costs credibility.

3 Fix by category, not by finding

Take the report's top contributor, fix its cause, re-export, re-run. Repeat while the score moves by more than a point or two. Do not work down a list of 3,000 elements.

4 Compare against the last delivery

A run diff answers the only question a reviewer really has: what changed? New findings are the story. Resolved findings are the receipt.

5 Federate before you promise coordination

A file that is clean alone can still be unusable together — coordinates, units and storey structure only misbehave in company.

6 Assemble the evidence pack

Chapter 6. The report, the record, the issues, the asset data. Ten minutes, and it is what the next four emails would have been.

7 Transmit with an explicit statement of suitability

Name the revision, the status, the score, the rule set and any waived findings. Chapter 7 has the template.

Step 3 in detail: fix the cause, not the symptom

The score model of chapter 3 was designed to make this the obvious move. Because repeats are logarithmic, a rule that fires 4,000 times and a rule that fires 40 times can

carry comparable penalties. Sorting by *penalty contribution* rather than by *finding count* puts the biggest lever at the top of the list — and the biggest lever is almost never "edit 4,000 elements".

WHAT THE REPORT SAYS	WHAT IT USUALLY MEANS	WHERE THE FIX LIVES
Thousands of elements missing a property set	The export template does not include that pset	Export configuration
Everything orphaned from the spatial structure	Elements are not assigned to levels, or the exporter was told to ignore them	Model organisation + export configuration
Every GUID changed since last revision	The model was recreated, copied, or exported through a round trip	Process, not file — see the caution below
A handful of individually broken elements	Genuine modelling mistakes	The model

The one fault you cannot fix after the fact

If GUIDs change on every export, nothing downstream can track anything: issues detach, revision comparison becomes meaningless, and asset data cannot be reconciled. This is a process defect. Fix it at the source, not with a find-and-replace — rewriting identifiers on delivery is worse than the original problem, because it looks stable and is not.

Step 4 in detail: the run diff

Comparing two validation runs of the same model at two revisions produces three lists, and each one has an audience:

- **New** — findings that were not there last time. This is a regression report. If the number is not zero, say why in the transmittal.
- **Resolved** — findings that are gone. This is your evidence that the last review was acted on. It is also the only politically useful list in the set.
- **Persisting** — findings present in both. If they persist deliberately, they are waivers and belong in writing (chapter 7). If they persist accidentally, the review process is not working.

Step 5 in detail: what federation adds

Three whole classes of defect are invisible until models meet:

1 **Coordinate divergence**

Two models, each internally consistent, referenced to different origins. Both check clean. Federated, they are hundreds of metres apart. The coordinate-offset check catches the common cause — a model authored far from its local origin — but only agreement on a shared reference point prevents it.

2 **Unit mismatch**

One model in millimetres, one in feet. Some tools convert silently, some do not, and the failure is measured in months when it reaches a fabricator. Length units are checked per file for exactly this reason.

3 **Storey structure mismatch**

Same building, different level names and elevations per discipline. Every filter, every sheet, every issue that says "level 3" now needs a translation table maintained by a human.

A note on file size

Large files are not a defect in themselves, but they correlate with two things worth knowing about: geometry exported at a level of detail nobody asked for, and elements exported as generic proxies instead of typed objects. Both are picked up by checks in chapter 4 — file-size anomaly and proxy overuse — precisely because the size is the symptom and the export configuration is the disease.

Before you spend a day optimising a model, check the cheap causes in this order: proxy elements, exported 2D and annotation content, unnecessary property sets on every element, and geometry detail on repeated components.

Where to spend the automation budget

If you automate exactly one thing in this workflow, automate step 2 — the check itself, on every export, whether or not it is going anywhere. A check that only runs before deliveries is a check that runs when the schedule is tightest and the appetite for bad news is lowest.

The evidence pack

06

A delivery is a claim: *this model is fit for this purpose at this revision*. Evidence is what turns the claim into something the receiver can check without repeating your work — and what stops the same argument happening twice.

Five artefacts, four of which you already have

ARTEFACT	ANSWERS	AUDIENCE
Validation report	What was checked and what was found	The receiving coordinator
Check record	That the check happened, on this exact file, at this time	The information manager, the client, an auditor
Issue file (BCF)	What is not fixed, and where to look	The person who has to act
Asset data (COBie)	What the building contains, as data	The client and the FM team
Transmittal note	Revision, status, score, rule set, waivers	Everyone, and the record

The validation report

Export it, attach it, do not paraphrase it. Three properties make a report useful to somebody who was not there:

- It names the **rule set** used, not just the results. A score without its rule set is uninterpretable.
- It states **coverage** — which rules ran, which did not (chapter 3). A report that cannot distinguish "passed" from "not attempted" is a marketing document.
- It carries **element identifiers**, so a finding can be located rather than searched for.

The check record

The weakest link in every quality process is that the check and the claim are separate. Anyone can say a model scored 92. A check record ties the number to a specific file: the file's own fingerprint, the rule set, the schema, the timestamp, the score.

Two properties make such a record worth more than a screenshot:

1 It is about the file, not the story

It should be derived from the file's content, so that changing a single byte of the model and re-issuing the same record is detectable.

2 It is verifiable by the receiver

Independently, without your involvement, and ideally without your tool. A record only your own software can confirm is a promise, not evidence.

Why this matters more each year

As soon as a delivery is a payment milestone — and increasingly it is — "we checked it" stops being a professional courtesy and starts being a claim somebody may contest. The cheapest moment to make it verifiable is before it is contested.

The issue file

BCF exists so that issues survive leaving your screen. Its whole value is the viewpoint: a camera position, a selection, and a comment, which together mean the receiver spends ten seconds finding the problem instead of ten minutes.

Two conventions make the difference between a BCF file that gets acted on and one that gets ignored:

- One topic per **cause**, not per element. Four thousand elements missing a pset are one topic with a representative viewpoint, not four thousand topics.
- Title the topic with the rule identifier and the fix, not the symptom.

`RULE_MISSING_PROPERTY_SET – add Pset_WallCommon to the export template`
is actionable. "Missing properties" is not.

Asset data

COBie is the point where model quality becomes commercially visible, because it is delivered to somebody who never opens a model and has no way to interpret an excuse. It is also, usefully, a validator of your validator: spaces without names, types without manufacturers, components without spaces — those gaps show up as blank columns that a facilities manager can see at a glance.

If a COBie export from your model is embarrassing, the model is not ready, whatever the geometry looks like.

The transmittal note

Six lines, and it prevents most delivery disputes:

TRANSMITTAL TEMPLATE

IFC delivery — statement of suitability

Container: {filename}

Revision / status: {rev} · {suitability code}

Schema: {IFC2X3 | IFC4 | IFC4X3}

Checked: {date} · rule set {name/version} · {n} of {n} checks completed

Health Score: {score}/100

Known open findings (agreed): {rule id — reason — agreed with — date}

Not suitable for: {e.g. quantity take-off, fabrication}

The last line is the one people skip, and it is the one that protects you. Stating what a container is *not* for is not defensive — it is the definition of a level of information need, delivered at the only moment anybody reads it.

The contract layer

07

Everything so far assumes somebody agreed what "good" means. Usually nobody did — which is why quality arguments on projects are so bitter and so unresolvable. This chapter is the paperwork that ends them, written to be copied.

The clauses below are drafting aids, not legal advice: adapt them to your project, your appointment and your jurisdiction, and have them reviewed by whoever signs your contracts.

Four clauses for the BEP

These belong in the BIM Execution Plan, in the information delivery section, and they are deliberately short. A quality clause nobody reads has no effect; a quality clause that fits on half a page gets quoted back at people, which is the point.

CLAUSE 1 Automated check before issue

Every IFC container issued to the CDE at status S2 (Shared) or above shall have been checked with the project's agreed rule set within the 24 hours preceding issue. The check report shall be issued alongside the container. Containers issued without a check report may be rejected without review.

CLAUSE 2 Minimum quality threshold

IFC containers shall achieve a Health Score of at least 80/100 under the project rule set. Containers scoring below the threshold may be issued only with the prior written agreement of the Information Manager, recording the findings concerned and the reason.

CLAUSE 3 Identifier stability

IFC GlobalIds shall be persistent for the life of the project: the identifier of an element shall not change between revisions unless the element itself is deleted and replaced. Task teams shall configure authoring and export tools accordingly, and shall report any event that invalidates identifiers (model recreation, round-trip import, template migration) at the time it occurs.

CLAUSE 4 Shared coordinates and units

All task teams shall use the project shared reference point and rotation defined in {document}, and shall deliver in metric SI length units. Storey names and elevations shall follow the agreed level schedule without local variation.

If you can only add one clause

Add clause 3. Threshold clauses improve the average delivery; the identifier clause is the only one that prevents a class of damage that cannot be repaired later.

The acceptance criteria table

Paste this into the EIR or the BEP appendix and fill in the right-hand column with your project's positions. Its purpose is not to be strict — it is to be *decided*, before the first delivery rather than during the third argument.

CRITERION	REJECT IF...	PROJECT POSITION
Structural integrity	Any schema or spatial finding at severity error	☐ Reject ☐ Accept with note
Health Score	Below the agreed threshold	Threshold: _____ /100
Check coverage	Any rule reported not-run or failed	☐ Reject ☐ Re-run required
Identifier stability	GUID turnover above an agreed percentage between revisions	Max turnover: _____ %
Naming	Filename does not follow the agreed convention	☐ Reject ☐ Rename and log
Georeferencing	Model not on the project shared reference point	☐ Reject
Units	Non-metric length units	☐ Reject
Classification	Elements without a classification reference	Applies from stage: _____
Property sets	Required psets missing for the declared level of information need	Pset schedule: _____
Spaces	Spaces without name, long name or floor area	Applies from stage: _____

From EIR text to checks

EIRs are written in prose by people who will never run a checker. Translating that prose into checks is a five-minute exercise that turns an aspiration into a gate. The pattern:

TYPICAL EIR SENTENCE	THE CHECK IT IMPLIES
“All elements shall be classified in accordance with {system}.”	Missing classification
“Models shall be delivered on the project coordinate system.”	Large coordinate offset + shared reference point agreement
“All spaces shall be named and measurable.”	Empty long name · space missing floor area
“Element identifiers shall be persistent.”	Duplicate GUID · invalid GUID format · GUID turnover between revisions
“Models shall be structured by building and storey.”	Orphan element · spatial hierarchy · missing storey · element in building
“Deliverables shall support asset information handover.”	Missing property set · missing type · missing material · space area

Each right-hand entry corresponds to one or more checks in chapter 4. Record the mapping in the BEP so that “compliant with the EIR” has an operational definition.

Two emails you will need

Both are written to be short, factual and impossible to take personally — which is the only register that works when the person on the other end is behind schedule.

EMAIL 1 Returning a container that cannot be accepted

Hi {name},

We've run the agreed pre-acceptance check on {filename} (rev {n}) and it comes back at {score}/100, below the {threshold} we set in clause {x} of the BEP.

The two findings driving that are:

- {rule id} — {plain description} ({n} elements)
- {rule id} — {plain description} ({n} elements)

Both look like export settings rather than modelling, so they should be quick — the report is attached with the element references.

We'll hold coordination on this container until the next issue. Happy to jump on a call if it's easier to look at the export configuration together.

{signature}

EMAIL 2 Issuing a container with known, agreed findings

Hi {name},

{filename} rev {n} is on the CDE at {status}, checked today: {score}/100 with all {n} checks completed.

Two findings remain open by agreement:

- {rule id} — outside the level of information need for this stage (agreed with {name}, {date})
- {rule id} — pending {supplier} information, expected {date}

Suitable for coordination. Not suitable for quantity take-off until the second item closes.

Check report attached.

{signature}

What these two emails have in common

Neither contains an adjective about the model. Both contain a number, a rule identifier, a cause and a date. That is the entire difference between a quality process and a quality opinion.

The pre-delivery checklist

Print it, or copy it into your issue template. Everything on it is either a check from chapter 4 or a step from chapter 5.

Before you export

- ☐ Using the project's named export configuration, not the default
- ☐ Elements assigned to the correct levels in the authoring model
- ☐ Shared reference point / project base point matches the project agreement
- ☐ Length units are metric SI
- ☐ Required property sets included in the export template

After you export, before it leaves your machine

- ☐ Validation run completed — all checks report as run, none failed
- ☐ Health Score at or above the project threshold
- ☐ Zero findings at severity error in the schema and spatial categories
- ☐ GUIDs stable against the previous revision
- ☐ Run diff reviewed: no unexplained new findings
- ☐ Storeys present, correctly named, in ascending elevation order
- ☐ Spaces named and carrying a floor area (if in scope for this stage)
- ☐ Classification references present (if in scope for this stage)
- ☐ File size in line with the previous revision, or the reason understood

Federated

- ☐ Model lands in the right place against the other disciplines
- ☐ Storey names and elevations align with the project level schedule
- ☐ No unit mismatch against the federated set

Transmittal

- ☐ Filename follows the project convention
- ☐ Revision and suitability status stated explicitly
- ☐ Validation report attached
- ☐ Check record attached (if used on the project)
- ☐ Open findings listed with reason, agreement and date
- ☐ What the container is NOT suitable for, stated in one line

Check index

All 44 checks, alphabetically by identifier, with category and default severity. Use it to look up a rule identifier you have been sent, or to build your project's severity overrides in one pass.

IDENTIFIER	CHECK	CATEGORY	DEFAULT
BROKEN_AGGREGATE	Broken aggregate	IFC Schema	ERROR
CIRCULAR_REFERENCE	Circular reference	IFC Schema	ERROR
CLASH_MEP_STRUCTURAL	MEP/structural clash	Clash detection	WARNING
CONNECTED_MEP	Disconnected MEP segment	MEP	WARNING
COORDINATE_OFFSET	Large coordinate offset	Data quality	WARNING
DUPLICATE_GUID	Duplicate GUID	IFC Schema	ERROR
DUPLICATE_NAME	Duplicate name	Data quality	WARNING
ELEMENT_CLASH	Element clash	Clash detection	WARNING
ELEMENT_IN_BUILDING	Element in building	Spatial structure	WARNING
EMPTY_LONGNAME	Empty long name	Data quality	WARNING
EMPTY_NAME	Empty name	Data quality	ERROR
EMPTY_PROPERTY_VALUE	Empty property value	Data quality	WARNING
EMPTY_STOREY	Empty storey	Spatial structure	INFO
FILE_AUTHOR_MISSING	No file author	IFC Schema	INFO
FILE_DESCRIPTION_MISSING	No file description	IFC Schema	INFO
FILE_SIZE_ANOMALY	File size anomaly	Data quality	INFO
INVALID_GUID_FORMAT	Invalid GUID format	IFC Schema	ERROR
INVALID_IFC_VERSION	Invalid IFC version	IFC Schema	INFO
ISO19650_AUTHOR_INFO	ISO 19650 author info	ISO 19650	INFO
ISO19650_FILENAME	ISO 19650 filename	ISO 19650	WARNING
ISO19650_PROJECT_INFO	ISO 19650 project info	ISO 19650	WARNING

IDENTIFIER	CHECK	CATEGORY	DEFAULT
LOD_MATERIAL_LAYER_MISSING	No material layers	LOD / LOIN	WARNING
LOD_PSET_MISSING	LOD Pset missing	LOD / LOIN	WARNING
LOD_QUANTITY_MISSING	LOD quantities missing	LOD / LOIN	WARNING
MEP_SYSTEM_MISSING	MEP system missing	MEP	WARNING
MISSING_BUILDING	Missing building	Spatial structure	WARNING
MISSING_CLASSIFICATION	Missing classification	Classification	WARNING
MISSING_MATERIAL	Missing material	Data quality	WARNING
MISSING_PROJECT	Missing project	IFC Schema	ERROR
MISSING_PROPERTY_SET	Missing property set	Data quality	WARNING
MISSING_STOREY	Missing storey	Spatial structure	WARNING
MISSING_TYPE	Missing type	Data quality	INFO
NAMING_CONVENTION	Naming convention	Data quality	WARNING
OPENING_WITHOUT_HOST	Opening without host	Spatial structure	WARNING
ORPHAN_ELEMENT	Orphan element	Spatial structure	ERROR
PROJECT_LONGNAME_MISSING	Project has no long name	Data quality	WARNING
PROXY_OVERUSE	Proxy overuse	Data quality	WARNING
SPACE_AREA_MISSING	Space missing floor area	Data quality	WARNING
SPATIAL_HIERARCHY	Spatial hierarchy	Spatial structure	ERROR
STOREY_ELEVATION_DUPLICATE	Duplicate storey elevation	Spatial structure	WARNING

IDENTIFIER	CHECK	CATEGORY	DEFAULT
STOREY_ELEVATION_MISSING	Storey elevation missing	Spatial structure	WARNING
STOREY_ELEVATION_ORDER	Storey elevation out of order	Spatial structure	WARNING
UNIT_CONSISTENCY	Imperial length unit	IFC Schema	WARNING
WRONG_CONTAINER	Wrong container	Spatial structure	ERROR

Glossary

Terms used in this handbook, defined the way a coordinator uses them rather than the way a standard writes them.

TERM	WORKING DEFINITION
BCF	BIM Collaboration Format. An exchange file for issues: each topic carries a comment and a viewpoint (camera + selection) so the receiver lands on the problem.
BEP	BIM Execution Plan. Where the project writes down how information will be produced and delivered — including, if you take chapter 7 seriously, what “acceptable” means.
CDE	Common Data Environment. The single agreed place where information containers live, with their revision and suitability status.
COBie	A structured schedule of the asset data in a model — spaces, types, components, systems — delivered to the client and the facilities team.
EIR	Exchange Information Requirements. What the client asks for, in prose. Chapter 7 turns it into checks.
Federation	Combining discipline models into one coordinated whole without merging them. Where coordinate, unit and storey mismatches surface.
GlobalId (GUID)	The 22-character identifier that makes an element the same element across revisions and tools. Its stability is the foundation of everything downstream.
Health Score	A 0–100 compression of a validation run, weighted by category and severity, with logarithmically diminishing repeats. See chapter 3.
IFC	Industry Foundation Classes. The open, vendor-neutral schema for exchanging building information. IFC2X3 and IFC4 are the two you will meet most.
Information container	ISO 19650's unit of delivery: a named, versioned, status-carrying file — your IFC, your drawing set, your COBie sheet.
Level of information need	How much detail — geometric, alphanumeric, documentary — a purpose actually requires. Over-delivery is a defect.
Property set (Pset)	A named group of properties attached to an element. Where almost all non-geometric information in an IFC lives.
Proxy	A generic IFC object used when the exporter cannot map an element to a real IFC class. Renders fine; carries no meaning.

TERM	WORKING DEFINITION
Run diff	A comparison of two validation runs: new, resolved and persisting findings. The core of a delivery review.
Spatial structure	The project → site → building → storey → space hierarchy that every element should hang from. Break it and filtering, scheduling and federation degrade.
Suitability	What the receiver is permitted to do with a container at its current status — look, coordinate, build.
Waiver	A finding accepted deliberately, with a reason, an owner and a date. The opposite of a finding ignored.

Where the settings live

A map, not a manual: the place in each authoring tool where most of chapter 4 is decided. Exact menu paths move between versions, so these are named by function. The per-check instructions in chapter 4 are the detailed version.

TOOL	WHERE THE EXPORT IS CONFIGURED	THE SETTINGS THAT FIX THE MOST FINDINGS
Revit	IFC export dialog → Modify Setup (saved setups can be exported and shared as a project standard)	Property sets to export · IFC class mapping table · level/storey handling · coordinate base · schema version
ArchiCAD	IFC Translators (saved translator per exchange purpose)	Property mapping · classification mapping · geometry conversion · storey and zone handling
Tekla Structures	IFC export dialog + user-defined attribute mapping	Numbering (identifier stability) · attribute-to-property mapping · assembly vs part export · base point
Allplan	IFC export configuration + attribute assignment	Attribute mapping to IFC properties · building structure assignment · export schema

In every one of these tools the export configuration is a shareable artefact. Store it with the project template and reference it by name in the BEP — that single move is worth more than any individual setting.

The end

If this handbook changes one thing about how you work, make it this: run the check *before* the file leaves your machine, every time, and attach what it produced. Everything else in these pages is detail hanging off that one habit.